Solid Waste Management

Debasree Purkayastha
School of Infrastructure, IIT Bhubaneswar
Email id: debasreep@iitbbs.ac.in

What is “Waste”?

Waste is a product or substance which is no longer suited for its intended use.

1.05 billion tonnes
132 kg/person/year

783 Million People
SUFFER FROM HUNGER

400 million metric tons per year

Source: 1. University of Tartu, MOOC: Auditing waste management, Accessed on: 19 December, 2024, https://sisu.ut.ee/waste/11-definition-and-classification-waste/, 2. Food Waste Index Report 2024, UN Environment Program, https://www.unep.org/resources/publication/food-waste-index-report-2024

Types of Solid waste

- **Solid wastes** are wastes that aren't liquid or gaseous, such as durable goods, nondurable goods, containers and packaging, food scraps, yard trimmings, and miscellaneous inorganic wastes. Solid waste is more or less synonymous with the term refuse, but solid waste is the preferred term.

- **Municipal solid waste (MSW)** is solid waste from residential, commercial, institutional, and industrial sources, but it does not include such things as construction waste, automobile bodies, municipal sludge, combustion ash, and industrial process wastes even though those wastes might also be disposed of in municipal waste landfills or incinerators.

- **Industrial wastes** are those wastes arising from industrial activities and typically include rubbish, ashes, demolition and construction wastes and special wastes.

- **Hazardous Waste:** Wastes that pose a substantial danger immediately or over a period of time to human, plant, or animal life are classified as hazardous wastes. A waste is classified as hazardous if it exhibits any of the following characteristics: (1) ignitability. (2) corrosivity, (3) reactivity, or (4) toxicity. The chemical category includes wastes that are corrosive, reactive, or toxic. The principal sources of hazardous biological wastes are hospitals and biological research facilities.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Gilbert M. Masters et al. Introduction to Environmental engineering and science (2008)

Types of Solid waste

Municipal Waste

Industrial Waste

Hazardous Waste

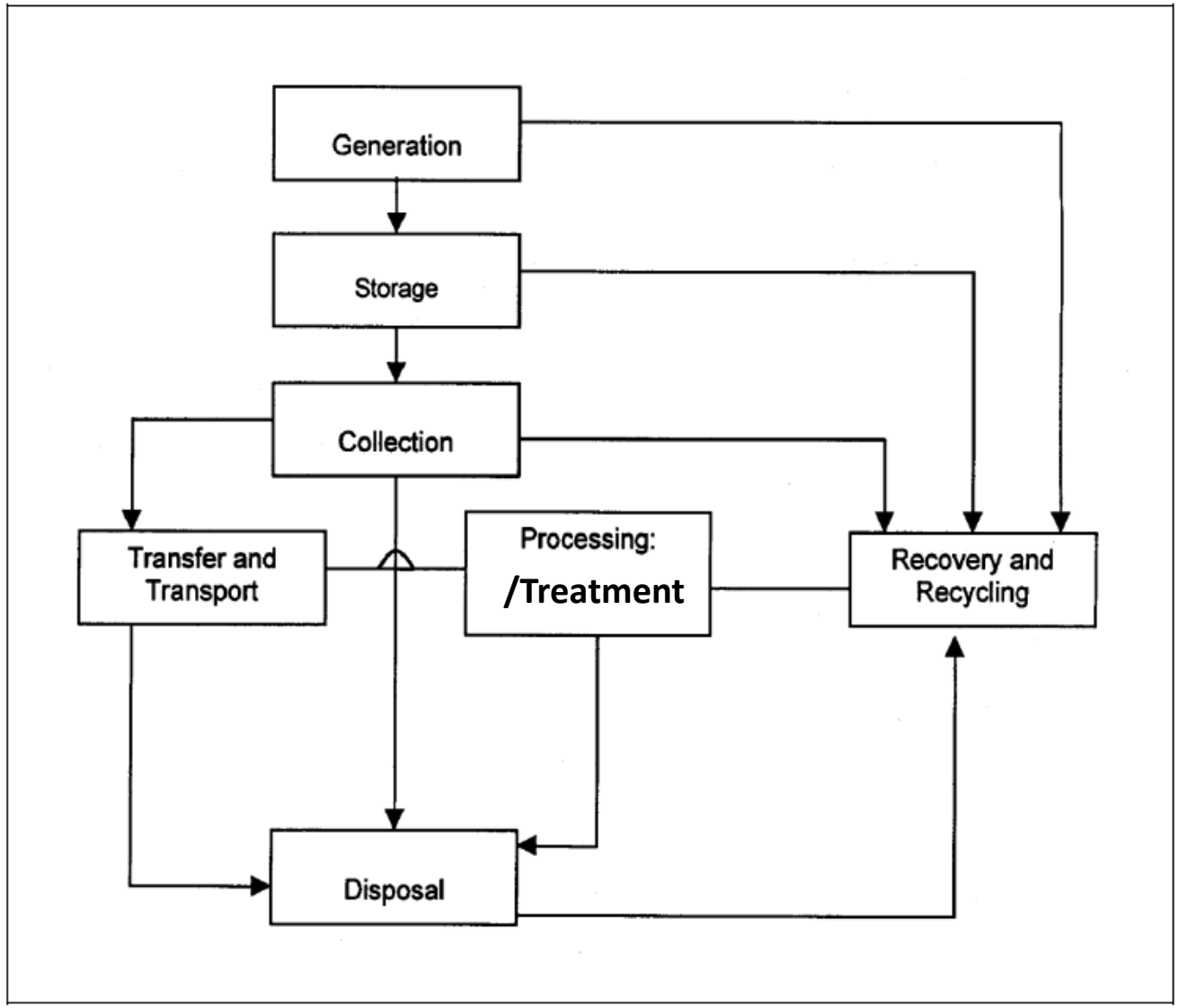
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Solid Waste

Component	Description
Food waste	The animal, fruit, or vegetable residues (also called garbage) resulting from the handling, preparation, cooking, and eating of foods. Because food wastes are putrescible, they will decompose rapidly, especially in warm weather.
Rubbish	Combustible and noncombustible solid wastes, excluding food wastes or other putrescible materials. Typically, combustible rubbish consists of materials such as paper, cardboard, plastics, textiles, rubber, leather, wood, furniture, and garden trimmings. Noncombustible rubbish consists of items such as glass, crockery, tin cans, aluminum cans, ferrous and nonferrous metals, dirt, and construction wastes.
Ashes and Residues	Materials remaining from the burning of wood, coal, coke, and other combustible wastes. Residues from power plants normally are not included in this category. Ashes and residues are normally composed of fine. powdery materials, cinders, clinkers, and small amounts of burned and partially burned materials.
Demolition and Construction waste	Wastes from razed buildings and other structures are classified as demolition wastes. Wastes from the construction, remodeling, and repairing of residential, commercial, and industrial buildings and similar structures are classified as construction wastes. These wastes may include dirt, stones, concrete, bricks, plaster, lumber, shingles, and plumbing, heating, and electrical parts.
Special waste	Wastes such as street sweepings, roadside litter, catch-basin debris, dead animals, and abandoned vehicles are classified as special wastes.
Treatment plant waste	The solid and semisolid wastes from water, wastewater, and industrial-waste treatment facilities are included in this classification.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Solid waste Management



Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

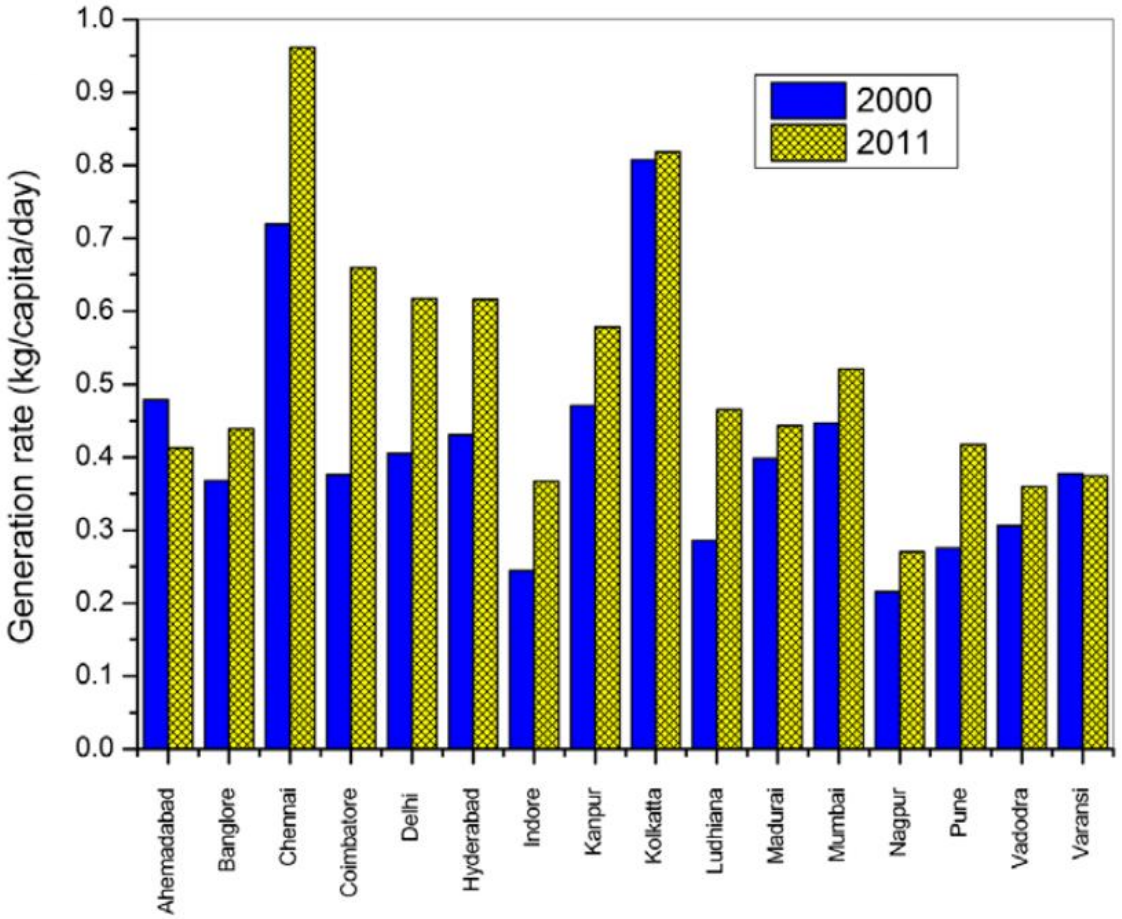
Waste generation: India

Calculated and expected population growth and national waste generation rate

Year	Urban population (million)	Per capita waste generation rate (kg/day)	National waste generation rate (million tons/year)
2001	197.3	0.439	31.63
2011	260.1	0.498	47.3
2021	342.8	0.569	71.15
2031	451.8	0.649	107.01
2036	518.6	0.693	131.24
2041	595.4	0.741	160.96

Area of land required for unsanitary disposal of municipal solid waste generated in India

Years	Area of land occupied/required for MSW disposal (km^2)	City equivalents (areawise)
1947–2001	240	50% of Mumbai
1947–2011	380	90% of Chennai
1947–2021	590	Hyderabad
2009–2047	1,400	Hyderabad + Mumbai + Chennai



Per capita generation of MSW selected Indian cities in 2000 and 2011

Source: 1. Singh, Rajeev Pratap, et al. "An overview for exploring the possibilities of energy generation from municipal solid waste (MSW) in Indian scenario." Renewable and Sustainable Energy Reviews 15.9 (2011): 4797-4808.

Physical Composition of Solid Waste

INDIA

Year	Biodegradables	Paper	Plastic/rubber	Metal	Glass	Rags	Others	Inert
1996	42.21	3.63	0.60	0.49	0.60	–	–	45.13
2005	47.43	8.13	9.22	0.50	1.01	4.49	4.02	25.16
2011	42.51	9.63	10.11	0.63	0.96	–	–	17.00

Source: Planning Commission Report.

- **Biodegradable Waste**: Food and kitchen waste, green waste (vegetables, flowers, leaves, fruits) and paper.
- **Recyclable Material**: Paper, glass, bottles, cans, metals, certain plastics, etc.
- **Inert Waste Matter**: C&D, dirt, debris.
- **Composite waste**: Waste clothing, Tetra packs, waste plastics such as toys.
- **Domestic Hazardous Waste** (also called "household hazardous waste") and toxic waste: Waste medicine, e-waste, paints, chemicals, light bulbs, fluorescent tubes, spray cans, fertilizer and pesticide containers, batteries, and shoe polish.

Source: I. Joshi, Rajkumar, and Sirajuddin Ahmed. "Status and challenges of municipal solid waste management in India: A review." Cogent environmental science 2.1 (2016): 1139434.

Physical Composition of Solid Waste

Characteristics	Low income	High income	Comments
Paper	1 – 4%	20 – 50%	Low paper content indicates low calorific value.
Plastics	1 – 6%	5 – 10%	Plastic is low as compared to high-income areas though the use of plastic has increased in recent years.
Ash and Fines	17 – 62%	3 – 10%	Ash and fines do not contribute to combustion
Moisture Content	30 – 40%	15 – 30%	Moisture content depends largely on the nature of the waste, climate and collection frequency. Waste can dry out while awaiting collection.
Bulk Density	300 – 400 kg/ m^3	150 kg/m^3	Heavier waste may cost more to handle and difficult to burn.

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Physical Composition of Solid Waste

Individual Components

Density

Particle Size

Moisture Content

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Physical Composition- Density

- Density of waste, i.e., its mass per unit volume (kg/m^3), is a critical factor in the design of a Solid Waste Management system, e.g., the design of sanitary landfills, storage, types of collection and transport vehicles, etc.

Components	Mass %		Moisture content %		Density in kg/m^3	
	Range	Typical	Range	Typical	Range	Typical
Food wastes	6-26	14	50-80	70	120-480	290
Paper	15-45	34	4-10	6	30-130	85
Cardboard	3-15	7	4-8	5	30-80	50
Plastics	2-8	5	1-4	2	30-130	65
Textiles	0-4	2	6-15	10	30-100	65
Rubber	0-1	0.5	1-4	2	90-200	130
Leather	0-2	0.5	8-12	10	90-260	160
Garden Trimming	0-20	12	30-80	60	60-225	105
Wood	1-4	2	15-40	20	120-320	240
Misc. Organic substances	0-5	2	10-60	25	90-360	240
Glass	4-16	8	1-4	2	160-480	195
Tin cans	2-8	6	2-4	3	45-160	90
Non-ferrous metals	0-1	1	2-4	2	60-240	160

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Physical Composition- Moisture Content

- Moisture content is defined as the ratio of the weight of water (wet weight - dry weight) to the total weight of the wet waste.
- Moisture increases the weight of solid wastes, and thereby, the cost of collection and transport.
- In addition, moisture content is a critical determinant in the economic feasibility of waste treatment by incineration, because wet waste consumes energy for evaporation of water and in raising the temperature of water vapour.

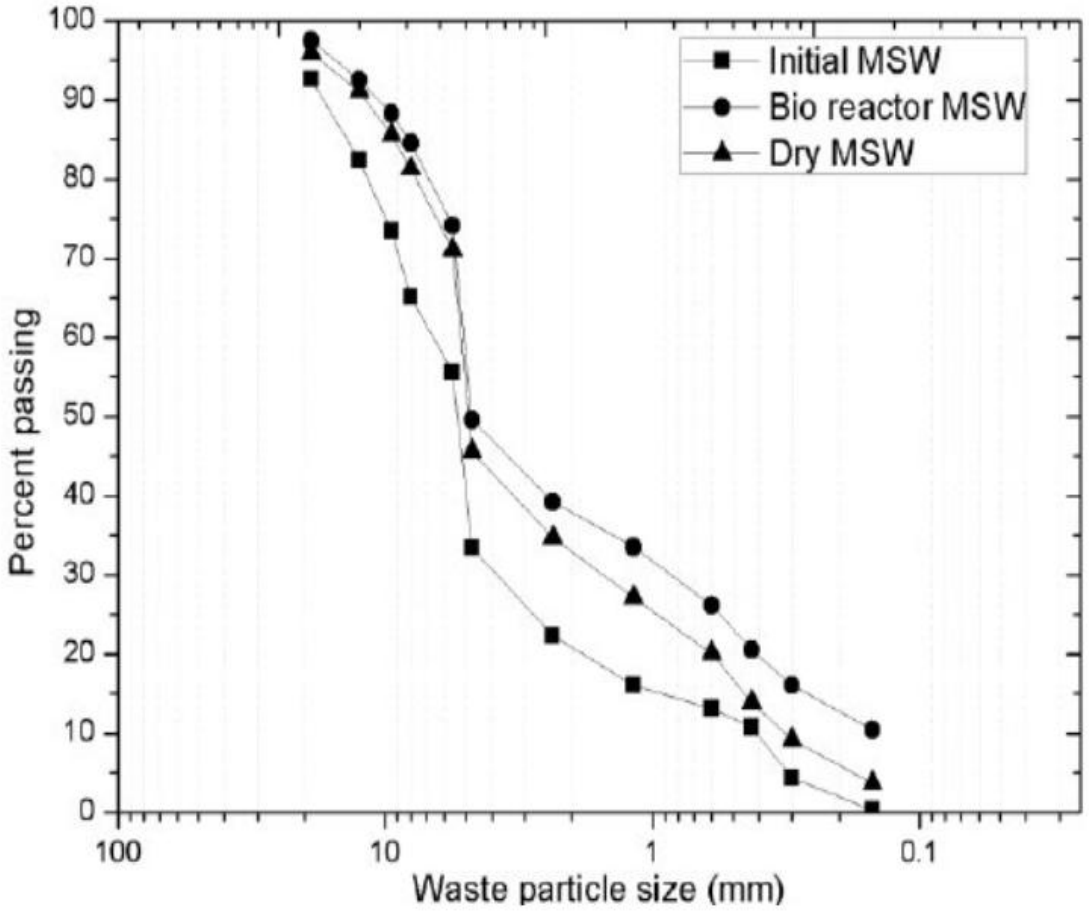
$$\text{Moisture content (\%)} = \frac{\text{Wet weight - Dry weight}}{\text{Wet weight}} \times 100$$

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Physical Composition- Particle size

- Measurement of size distribution of particles in waste stream is important because of its significance in the design of mechanical separators and shredders.

- Generally, the results of size distribution analysis are expressed in the manner used for soil particle analysis. That is to say, they are expressed as a plot of particle size (mm) against percentage, less than a given value.



Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005
2. Lakshmikanthan, P., L. G. Santhosh, and G. L. Sivakumar Babu. "Evaluation of bioreactor landfill as sustainable land disposal method." Sustainability Issues in Civil Engineering (2017): 243-254.

Numerical

Determine the moisture content of the sample from the following data:
*Based on a 100 kg sample of waste

Components	Percent by mass*	Moisture Content (%)
Food wastes	15	70
Paper	45	6
Cardboard	10	5
Plastics	10	2
Garden trimmings	10	60
Wood	5	20
Tin Cans	5	3

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Determine the moisture content of the sample from the following data:
*Based on a 100 kg sample of waste

Components	Percent by mass*	Moisture Content (%)	Dry mass (kg)
Food wastes	15	70	4.5
Paper	45	6	42.3
Cardboard	10	5	9.5
Plastics	10	2	9.8
Garden trimmings	10	60	4.0
Wood	5	20	4.0
Tin Cans	5	3	4.9
			79.0

$$Moisture\ Content = \frac{100 - 79}{100} x100 = 21\%$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Determine the density of the sample from the following data:
*Based on a 1000 kg sample of waste

Components	Percent by mass*	Typical density (kg/m^3)
Food wastes	15	290
Paper	45	85
Cardboard	10	50
Plastics	10	65
Garden trimmings	10	105
Wood	5	240
Tin Cans	5	90

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Determine the density of the sample from the following data:
*Based on a 1000 kg sample of waste

Components	Percent by mass*	Typical density (kg/m^3)	Volume (m^3)
Food wastes	15	290	0.52
Paper	45	85	5.29
Cardboard	10	50	2.00
Plastics	10	65	1.54
Garden trimmings	10	105	0.95
Wood	5	240	0.21
Tin Cans	5	90	0.56
			11.07

$$Density = \frac{1000}{11.07} = 90.33 \text{ kg/m}^3$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Chemical Composition of Solid Waste

Important in evaluating alternative processing and energy recover options.

- Proximate Analysis:
  - Moisture (Loss at 105°C for 24 h)
  - Volatile matter (additional loss on ignition at 550°C for 1-2 h)
  - Ash: Residue after burning
  - Fixed Carbon= VS %/1.8
- Ultimate Analysis: Percent of Carbon (C), Hydrogen (H), Nitrogen (N), Sulfur (S), Oxygen (O)
- Heating value (Calorific Value)

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Chemical Composition – Volatile solid

Step I

Weigh the empty crucible.

Step 2

Weigh the dried and/or sieved sample.

Step 3

Heat the sample in furnace at 550◦ C for 2 h

Step 4

Note the final weight of the sample and calculate the VS.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

Assignment

Question 1: Describe the principle and functioning of Bomb Calorimeter.

Question 2: What is the waste composition and waste generation rate of your city?

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Thank you

For any questions, please feel free to reach out to me.